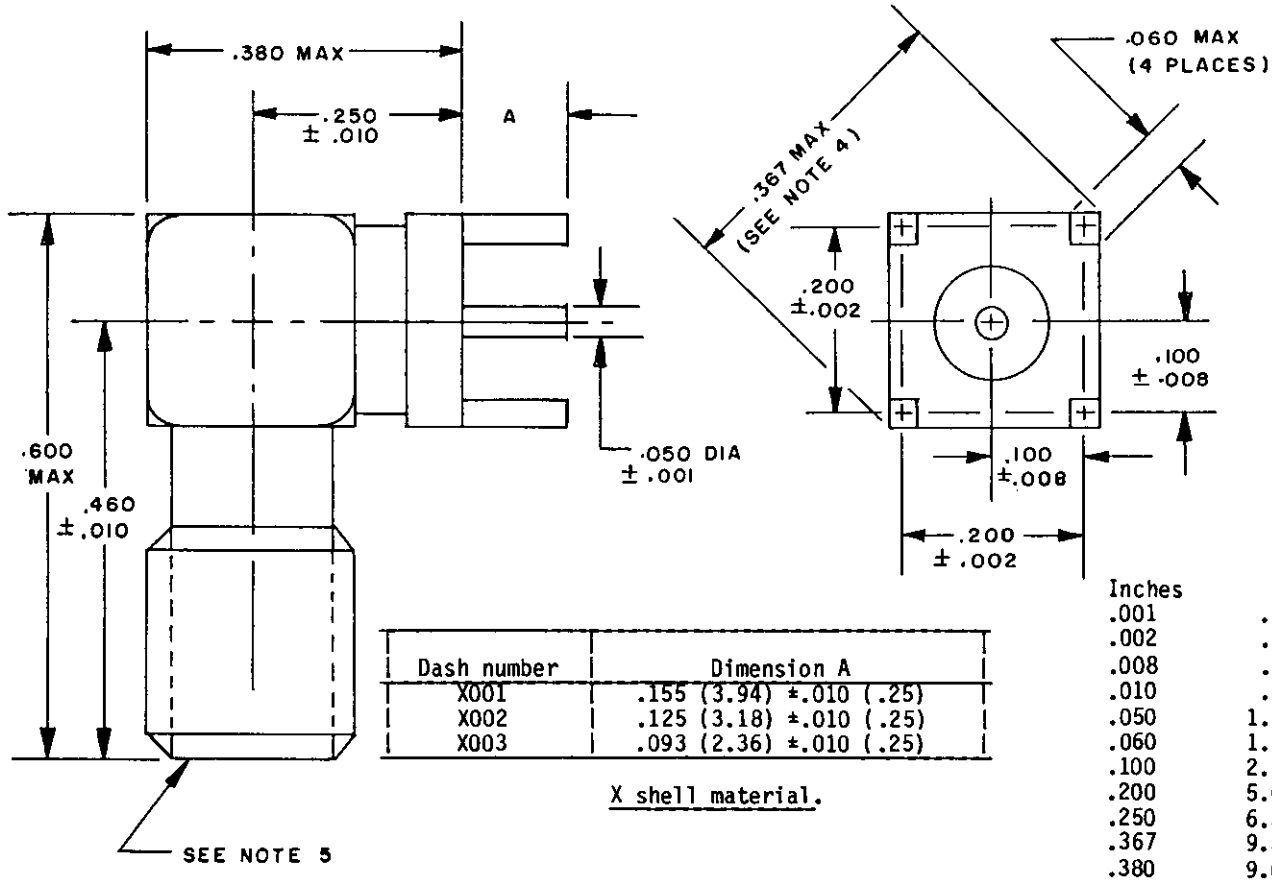
MILITARY SPECIFICATION SHEET

CONNECTORS, RECEPTACLE, ELECTRICAL, COAXIAL, RADIO FREQUENCY, SERIES SMA (UNCABLED, FEMALE, PRINTED CIRCUIT, RIGHT ANGLE, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-39012.



Dash number	Dimension A
X001	.155 (3.94) ±.010 (.25)
X002	.125 (3.18) ±.010 (.25)
X003	.093 (2.36) ±.010 (.25)

X shell material.

Inches	mm
.001	.02
.002	.05
.008	.20
.010	.25
.050	1.27
.060	1.52
.100	2.54
.200	5.08
.250	6.35
.367	9.32
.380	9.65
.460	11.68
.600	15.24

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All undimensioned pictorial configurations are for reference purposes only.
4. Configuration optional.
5. Series SMA, socket contact interface in accordance with MIL-STD-348.
6. All corrosion resistant steel bodied connectors shall be gold plated in accordance with MIL-G-45204, type II, Class I.

FIGURE 1. General configuration.

(A) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:12Z
Check the source to verify that this is the current version before use.

ENGINEERING DATA

Nominal impedance: 50 ohms.

Frequency range: .5 to 18 GHz.

Voltage rating: 335 V rms, maximum, at sea level.
85 V rms, maximum, at 70,000 feet.

Temperature rating: -65°C to +105°C.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Force to engage and disengage:
Longitudinal force - Not applicable.
Torque - 2 inch-pounds, maximum.

Coupling proof torque: Not applicable.

Inspection conditions: For each test of threaded coupling connector where the test is performed on mated pairs, the pairs shall be torqued to 7 to 10 inch-pounds.

(A) Mating characteristics (see figure 2 and MIL-STD-348 for dimensions):
Center contact (female):
Oversize test pin - .0375 + .0001
Test pin finish - 16 microinches.
Insertion depth - .030/045.
Number of insertions - 3.

Insertion force test: Steel test pin diameter .0370 + .0001.
Insertion depth - .050/.075.
Test pin finish - 16 microinches.
Insertion force - 3 lbs, maximum.

Withdrawal force test: Steel test pin diameter .0355 -.0001.
Insertion depth - .050/.075.
Withdrawal force - 1 ounce, minimum.
Test pin finish - 16 microinches.

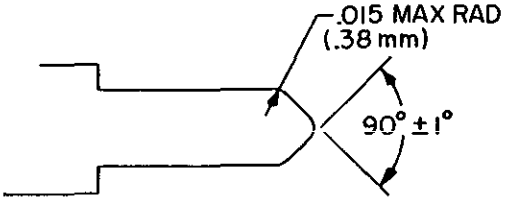


FIGURE 2. Test pin data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:12Z
Check the source to verify that this is the current version before use.

Hermetic seal: Not applicable.

Leakage (pressurized connectors): Not applicable.

Insulation resistance: Method 302, test condition B, MIL-STD-202.
5,000 megohms, minimum.

Center contact retention: 6 lbs minimum axial force. 4 inch-ounces, minimum, radial torque.

1/ Solderability: Method 208 of MIL-STD-202.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

Voltage standing wave ratio (VSWR): Not applicable.

Swept frequency VSWR test setup: Not applicable.

Connector durability: Insertion and withdrawal force.
500 cycles, minimum, at 12 cycles/min, maximum.
The connector shall meet mating characteristics and force to engage and disengage requirements.

Contact resistance: In milliohms, maximum.

	Initial	After environment
Center contact:	3.0	4.0
Outer contact:	2.0	Not applicable

Dielectric withstanding voltage at sea level: Method 301, MIL-STD-202.
1,000 V rms.

Vibration high frequency: Method 204, test condition D, MIL-STD-202.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.

Thermal shock: Method 107, test condition B, MIL-STD-202, except test high temperature shall be +125°C.

Moisture resistance: Method 106 of MIL-STD-202.
No measurements at high humidity. Insulation resistance shall be at least 200 megohms within 5 minutes after removal from humidity.

Corona level:
Voltage - 250 volts, minimum.
Altitude - 70,000 feet.

RF high potential withstanding voltage:
Voltage and frequency - 670 V rms at 5 MHz.
Leakage current - Not applicable.

Cable retention force: Not applicable.

Coupling mechanism retention force: Not applicable.

RF leakage: Not applicable.

Insertion loss: Not applicable.

Group qualification: See table I.

Part number: M39012/94-(and dash number from figure 1).

1/ For quality conformance inspection, the test shall be performed in group B following VSWR; inspection level shall be S-1 of MIL-STD-105 and AQL 10 percent defective.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:12Z
Check the source to verify that this is the current version before use.

TABLE I. Group qualification.

Group	Submission and qualification of any of the following connectors	Qualifies the following connectors
I	M39012/94-X001 M39012/94-X002 M39012/94-X003	M39012/94-X001 M39012/94-X002 M39012/94-X003

X Shell material.

NOTE: If a connector manufacturer produces a connector which meets all the requirements for two or more connector types (within the same series), the manufacturer may receive qualificaton approval for the tow or more connector types by qualifying the one connector. It is not necessary that such connectors be in the same group. Each connector, however, must be marked with its own appropriate part number.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activites:
Army - MI
Navy - SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AM, AT, ME
Navy - AS, MC, OS
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3573-02)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T17:12Z
Check the source to verify that this is the current version before use.